



Demand Estimation and Market Analysis: Air Fryers

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Air Fryer Market

→ We studied the “Air Fryer” market using data from Amazon purchases to allow us to infer markups, and unit costs
→ We verified the data contained 10 brands, with the years 2019-2023, and that the expected number of rows was 50

KEY FINDINGS:

→ The largest shares are from brands like Ninja, Instant Pot, and GoWISE USA over the time period
→ Cuisinart, followed by Oster, are the two most expensive brands, while Dash, followed by Ultrean, are the two least expensive brands
→ Ninja has consistently kept highest ratings over time, followed by Ultrean and Cosori.
→ Compact products tend to have the largest share in the market

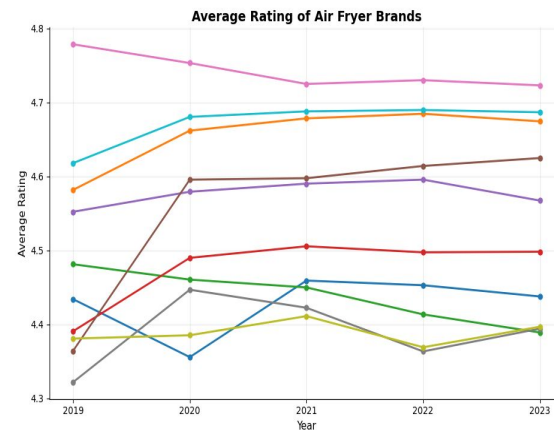
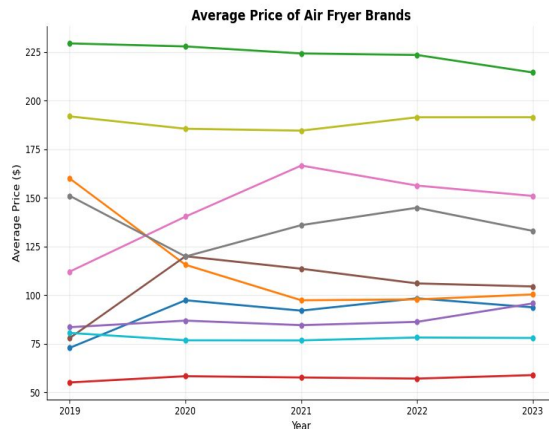
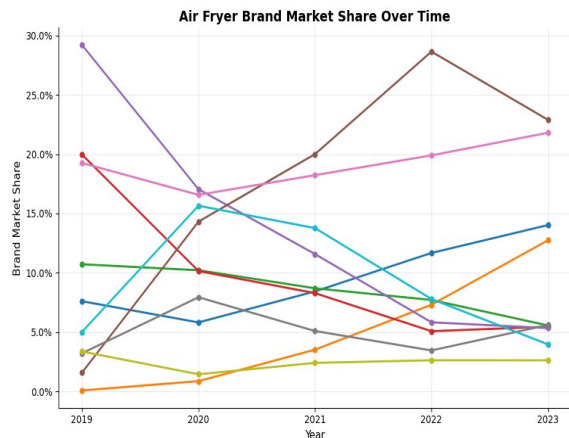
10 TOP BRANDS

DATA FROM
2019-2023

50 BRAND YEAR
OBSERVATIONS



Prices, Ratings, Market Shares



- Brand
- Chefman
 - Cosori
 - Cuisinart
 - Dash
 - GoWISE USA
 - Instant Pot
 - Ninja
 - NuWave
 - Oster
 - Ultrean

→ Overall, the market shows some stable leaders but also meaningful movement in brand shares across the five-year period



Demand Model

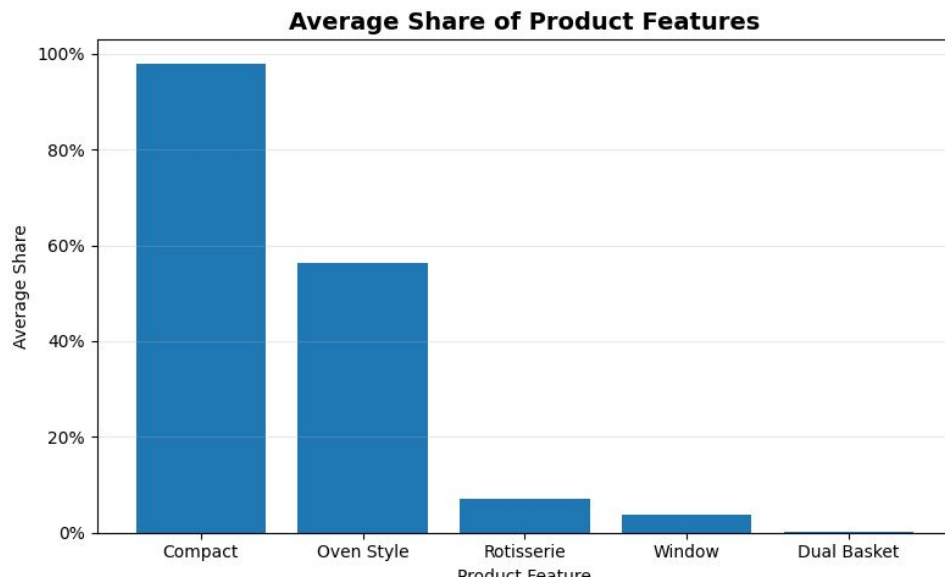
$$\log(\text{share_bt}) = \alpha_0 + \alpha_t (\text{year FE}) + \gamma_b (\text{brand FE}) + \beta_{\text{price}} \cdot \text{price} + \beta_{\text{rating}} \cdot \text{rating} + \sum \beta_l \cdot \text{features} + \varepsilon$$

- Price coefficient (β_{price}): -0.027
Negative γ means higher prices reduce demand
- R^2 : 0.891
Model explains 89% of log-share variation
- Year trend: +0.5
2021 shows strongest demand vs. 2019 baseline





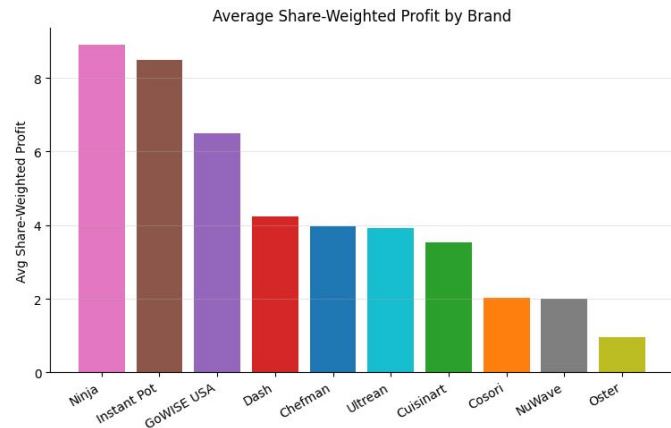
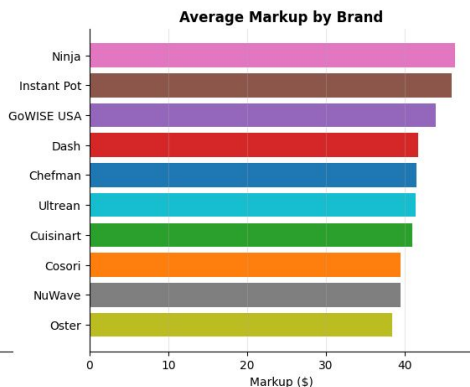
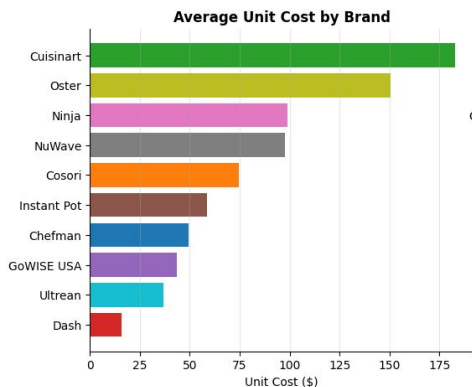
Product Features



→ Customers seem to prefer compact style air fryers
→ Less conventional products such as dual baskets and window styles are less popular



Estimated Unit Costs, Markups, Share-Weighted Profits



→ All brands charge nearly identical markups (\$38-\$47), so price differences across brands reflect costs rather than pricing power

→ **Cuisinart** and **Oster** have the highest unit costs but small shares, giving them the lowest profit index despite premium prices

→ **Ninja** and **Instant Pot** lead on share-weighted profit, driven by volume of sales rather than higher prices (selling more > charging more)

→ No negative unit costs were found, validating the demand model



Conclusions

→ Ninja is the brand dominating the market, as it holds the largest share (~22%), with a mid-high range price, and still has a positive brand fixed effect after controlling for price and features, meaning consumers actively prefer it beyond what price alone explains.

→ Instant Pot's shares appear to be driven by price and product characteristics rather than brand loyalty, as its brand coefficient is negative

→ GoWISE USA similarly wins on price but has been losing share over time, suggesting a weakening position

→ Cosori has been consistently growing through 2023, though it had smaller shares in previous years

→ Cuisinart and Oster occupy a distinct niche: modest market shares but the strongest brand fixed effects (+3.75 and +2.22 respectively), indicating a loyal, premium-oriented customer base that chooses them despite their higher prices. This is evidence of genuine brand equity, but untapped volume potential.

→ Markups are nearly identical across all brands (\$38-\$47), meaning competitive advantage comes from volume. Ninja and GoWISE USA win on profit through share, not pricing power. Cuisinart and Oster have genuine brand equity but leave profit on the table due to small shares.